

Leticia Mattos Da Silva

Massachusetts Institute of Technology
Room 32-D475A
32 Vassar Street, Cambridge, MA 02139

Email: leticiam@mit.edu

Website: lmattos.com

Education

Massachusetts Institute of Technology Ph.D. in Computer Science, advised by Justin Solomon	Sept. 2021 - <i>present</i>
Massachusetts Institute of Technology S.M. in Computer Science, advised by Justin Solomon	June 2023
University of California, Los Angeles B.S. in Mathematics <i>cum laude</i>	March 2021

Research Positions

Guest Researcher Flatiron Institute, Simons Foundation, New York, NY	Aug. 2024 - March 2025
Pre-Doctoral Researcher Flatiron Institute, Simons Foundation, New York, NY	May - Aug. 2024
Research Scientist Intern Adobe Inc., San Jose, CA (Remote)	May - Aug. 2022
Research Intern Fields Institute for Research in Mathematical Sciences, Toronto, ON (Remote)	July - Aug. 2021
Research Intern Fields Institute for Research in Mathematical Sciences, Toronto, ON (Remote)	July - Aug. 2020
Student Researcher SMALL Research Experience for Undergraduates, Williamstown, MA	June - Aug. 2019

Honors and Awards

WiGRAPH Rising Stars in Computer Graphics	2025
Oberwolfach Leibniz Graduate Student (OWLG)	2025
MathWorks Engineering Fellowship	2024
EECS MathWorks Fellowship	2023
Schwarzman College of Computing Fellowship funded by Google	2022
Google Lime Scholarship <i>semi-finalist</i>	2022
Distinguished Graduate Fellowship in Electrical Engineering and Computer Science	2021
Will Rogers Memorial Scholarship	2021
De Anza Physical Sciences, Mathematics, and Engineering Scholarship	2017

Publications

5. **Schrödinger Bridges on Discretized Geometric Domains** 2026
Leticia Mattos Da Silva, Mohammad Sina Nabizadeh, Justin Solomon
Conditionally accepted at SIGGRAPH 2026
4. **Variational Elastodynamic Simulation** 2025
Leticia Mattos Da Silva, Silvia Sellán, Natalia Pacheco-Tallaj, Justin Solomon
ACM SIGGRAPH 2025
3. **Mirror Bridges Between Probability Measures** 2025
Leticia Mattos Da Silva, Silvia Sellán, Francisco Vargas, Justin Solomon
Under Peer Review
2. **A Framework for Solving Parabolic Partial Differential Equations on Discrete Domains** 2024
Leticia Mattos Da Silva, Oded Stein, Justin Solomon
ACM Transactions on Graphics
1. **Breaking Good: Fracture Modes for Realtime Destruction** 2022
Silvia Sellán, Jack Luong, **Leticia Mattos Da Silva**, Aravind Ramakrishnan, Yuchuan Yang, Alec Jacobson
ACM Transactions on Graphics

Invited Talks

- Leveraging Latent Mathematical Structure in Geometry Processing** March - April 2026
Graphics Seminar, Boston University, *hosted by Ed Chien*
Geometric Computing Lab, NYU, *hosted by Daniele Panozzo*
Geometry and the City Lab, Columbia University, *hosted by Silvia Sellán*
PACM IDeAS Seminar, Princeton University, *hosted by Amit Singer*
Université de Montréal, *hosted by Mikhail Bessmeltsev*
- Schrödinger Bridges on Discrete Surfaces** Aug. 2025
Geometry Workshop in Obergurgl, Universitätszentrum Obergurgl
- A Framework for Solving Parabolic PDE on Discrete Domains** March 2025
Visgraf Seminar, Institute of Pure and Applied Mathematics (IMPA)
- Tackling Nonlinear and Nonconvex Challenges with Geometric Algorithms** Feb. 2025
Computer Graphics Seminar, ISTA, *hosted by Samara Ren*
- Geometric Algorithms, Stochasticity and the Fokker-Planck Equation** Feb. 2025
Oberwolfach Workshop *Mathematical Imaging and Surface Processing*
- The Differential Equations Multiverse: From PDE to SDE in Computer Graphics** Nov. 2024
Computational Science and Engineering Community Seminar, MIT
- Towards Broader Parabolic PDE Solvers in Computer Graphics** Sept. 2024
PAB Seminar, The University of Edinburgh, *hosted by Amir Vaxman*
- An Iterative High-Order Method for PDE on Moving Surfaces** Aug. 2024
Geometric Computing Lab, NYU, *hosted by Daniele Panozzo*
- Variational Characterizations of Nonlinear PDE on Geometric Domains** May 2024
The Scientific AI Group, UT Austin, *hosted by Richard Tsai*

Contributed Talks

Variational Elastodynamic Simulation SIGGRAPH, Vancouver, BC, Canada	Aug. 2025
A Framework for Solving Parabolic PDE on Discrete Domains 31st Annual Fall Workshop on Computational Geometry, Tufts University	Nov. 2024
A Framework for Solving Parabolic PDE on Discrete Domains SIGGRAPH, Denver, CO	July 2024
Log of Heat Equation Graduate Women in Course 6 Research Summit, MIT	Nov. 2021
Fracture Harmonics Women in Math Research Night, University of California, Los Angeles	May 2021
When Bands Play in Random Matrix Theory Young Mathematicians Conference, Columbus, OH, with Charles Devlin VI	Aug. 2019

Other Presentations

Perron's Method for the Landau Equation <i>Poster</i> , ICMS Workshop, Edinburgh, UK	Sept. 2024
An Iterative High-Order Method for PDE on Moving Surfaces <i>Poster</i> , Flatiron Institute, Simons Foundation, New York, NY	Aug. 2024
Solving Parabolic PDE with MATLAB <i>Poster</i> , MathWorks Day, MIT	April 2024
Solving Parabolic PDE on Discrete Domains <i>Poster</i> , ICERM Workshop <i>PDEs and Geometry: Numerical Aspects</i> , Brown University	March 2024

Teaching

Convex Optimization in Computer Graphics Tutorial Course Instructor, Eurographics, Aachen, Germany	May 2026
Applied Numerical Algorithms Teaching Fellow, MIT	Fall 2025
Convex Relaxation Strategies for Graphics and Geometry Processing Tutorial Course Instructor, Symposium on Geometry Processing, Bilbao, Spain	July 2025
Progressive Mesh Compression Teaching Assistant, Summer Geometry Institute, MIT, <i>supervised by Paul Kry</i>	Summer 2023
First-Year Engineering Transition Bridge: Calculus Instructor, University of California, Los Angeles	Summer 2019, 2020
Introduction to Engineering Disciplines Teaching Assistant, University of California, Los Angeles	Fall 2019
Intermediate Algebra, Precalculus, Integrated Statistics, Others Teaching Assistant, De Anza College	Fall 2016 - Spring 2018

Mentoring

Reese Long , Undergraduate Researcher, UROP, MIT	Spring 2026
Ryan Dong , Undergraduate Researcher, UROP, MIT	Spring 2025
Ayyoob Berhane , Undergraduate Researcher, SuperUROP Scholar, MIT	AY 2024 - 2025
Akshata Tiwari , Undergraduate Researcher, UROP, MIT	Winter Break 2024
Francisco Unai Caja López , Summer Geometry Institute Fellow	Summer 2023
Anna Cole Percy , Summer Geometry Institute Fellow	Summer 2023
Matheus da Silva Araujo , Summer Geometry Institute Fellow	Summer 2023
Hossam Mohamed Saeed Gize , Summer Geometry Institute Fellow	Summer 2023
Dora Zhao , Graduate Application Mentee, <i>now at Stanford</i>	Fall 2022
Olga Guţan , Graduate Application Mentee, <i>now at CMU</i>	Fall 2021

Academic Service

Co-organizer of the Visual Computing Seminar MIT	Sept. 2025 - <i>present</i>
Member of the Graduate Student Advisory Group School of Engineering, MIT	Sept. 2024 - <i>present</i>
Co-Organizer of the New England Symposium on Graphics MIT	April 2025, 2026
Flipping Failure Advisory Board Teaching and Learning Lab, MIT	Sept. 2023 - May 2024
Organizer of EECS Culture: Today and Tomorrow Event Visit Days, EECS Department, MIT	March 2024
Graduate Applications Reviewer EECS Department, MIT	Jan. 2024
Executive Member of Diversity, Equity, and Inclusion EECS Graduate Student Association, MIT	Jan. 2023 - Dec. 2023
Disability Justice Council Representative Graduate Student Council, MIT	Sept. 2021 - June 2023
Executive Member of New Student Orientation EECS Graduate Student Association, MIT	Jan. 2022 - Dec. 2022
Student Volunteer Eurographics/ACM SIGGRAPH Symposium on Geometry Processing (SGP)	July 2021
Volunteer for Transfer Student Welcome Undergraduate Admission, University of California, Los Angeles	May 2021
Co-organizer of Tutor Training Workshops Math Performance Success Program, De Anza College	Sept. 2017 - June 2018

Outreach

Organizer of Session on Graduate School Applications

Summer 2021, 2023, 2024, 2025

Summer Geometry Initiative, MIT

Mentor for the Graduate Assistance Application Program (GAAP)

Fall 2021, 2022, 2024

EECS Department, MIT

Volunteer for Graduate Recruitment Effort

Nov. 2023

The Society of Hispanic Engineers (SHPE) National Convention, Salt Lake City, UT

Volunteer for the Student Transfer Outreach and Mentor Program (STOMP)

April 2021

Undergraduate Admission, University of California, Los Angeles

Co-organizer of Outreach Program Noches de Ciencias

Sept. 2019 - March 2020

Society of Latino Engineers and Scientists, University of California, Los Angeles